# Brandon Rush

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## Education

**University of California, Berkeley** — Berkeley, CA

*Bachelor of Arts in Data Science; GPA: 3.61* — *Expected May 2028*

- **Relevant Coursework:** Machine Learning (CS 189)*, Discrete Mathematics and Probability Theory (CS 70)*, Probability and Random Processes (EECS 126), Optimization Models (EECS 127), Data Structures (CS 61B), Principles and Techniques of Data Science (Data 100) **In progress*

## Experience

**Climformatics** — Summer 2026

*Machine Learning / Data Science Intern* — *Remote*

- Shipped a production fire-weather dashboard for **69 California locations** with **1–7 day forecasts**, automated ETL, forecast blending, and CI-gated deployments to DigitalOcean; the system has run daily since August 2026
- Developed an hourly NWP bias-correction model with TSMixer and PyTorch, achieving $\boldsymbol{r = 0.80}$**–0.81** out-of-sample temperature correlation across 69 locations
- Built a **69GB** Open-Meteo pipeline with automated caching and feature engineering to support model experimentation, training, and the live product

**UC Berkeley College of Computing, Data Science, and Society** — Fall 2026

*Data Science Peer Advisor* — *Berkeley, CA*

- Explain technical and academic concepts to students and collaborate with peer and major advisors on course planning, research opportunities, and student resources

## Technical Projects

**Optimal Wordle Solver** | *Python, NumPy, Information Theory* — Spring 2026

- Built a probabilistic solver that maximizes entropy and outperformed average human play by **11%** (**3.49** vs. 3.92 guesses)
- Evaluated **30M+** guess–answer pairs with mutual information to identify an optimal opening guess and extract **5.89 of 11.18 bits** per guess

**Bayesian Painter Style Modeling** | *Python, PyTorch, Bayesian Inference* — Spring 2026

- Modeled artistic style for **5 painters** and **3,100+ WikiArt images** using MLE and MAP estimation on 256-dimensional VAE latent vectors
- Applied Normal-Inverse-Wishart priors to stabilize covariance estimates in low-data regimes and improved held-out log-likelihood by $\mathbf{10^8}$

**WordNet** | *Java, Data Structures, Web Applications* — Fall 2025

- Built a Java backend for **150MB** of Google Ngram data with a custom time-series structure, web-server query handlers, and browser-based visualizations

**Cook County Housing Price Predictor** | *Python, Pandas, Scikit-Learn* — Fall 2025

- Developed a multivariate regression pipeline with one-hot encoding and log transformations, then audited predictions for regressive bias in property-tax assessments

## Leadership & Athletics

**Cal Track and Field** — 2024–Present

*NCAA Division I Student-Athlete* — *Berkeley, CA*

- Balance **20+ hours** of weekly training, competition travel, and a full technical course load, demonstrating ownership, discipline, and performance under pressure

## Technical Skills

**Languages**: Python, Java, R, SQL, MATLAB
**Machine Learning & Data**: PyTorch, TensorFlow, Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn
**Tools & Platforms**: Git, Docker, Linux, Jupyter, DigitalOcean, PostgreSQL